

LAZARD LEGACY MATH ACADEMY

Personal Finance Math Basics

Real-World Money Math — 25 Minutes

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"Dreams can come true. Better late than never."

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Section 1: Quick Lesson — Budgeting with the 50/30/20 Rule

What is the 50/30/20 Rule?

This is a simple budgeting framework that divides your after-tax (take-home) income into three categories:

- **50% → Needs:** Rent/mortgage, utilities, groceries, insurance, minimum debt payments, transportation to work.
- **30% → Wants:** Entertainment, dining out, subscriptions (Netflix, Spotify), hobbies, vacations, shopping.
- **20% → Savings & Debt:** Emergency fund, retirement savings, extra debt payments, investments.

Why it works: It's simple, flexible, and gives you permission to spend on wants while still saving.

Worked Examples:

Example 1:

Monthly take-home pay is \$3,000. How much should go to Needs?

Solution: Needs = 50% of income. $50\% \times \$3,000 = 0.50 \times \$3,000 = \mathbf{\$1,500}$ for needs.

Example 2:

Monthly income is \$4,200. Calculate amounts for Wants and Savings.

Solution: Wants = $30\% \times \$4,200 = \mathbf{\$1,260}$. Savings = $20\% \times \$4,200 = \mathbf{\$840}$.

Example 3:

You earn \$2,800/month. Rent=\$1,100, utilities=\$200, groceries=\$350. Are you within the 50% Needs budget?

Solution: Total needs = \$1,650. Budget for needs = $50\% \times \$2,800 = \$1,400$.
You are **\$250 OVER budget**.

Example 4:

Annual salary is \$48,000. What's the monthly 50/30/20 breakdown?

Solution: Monthly = $\$48,000 \div 12 = \$4,000$. Needs: **\$2,000**. Wants: **\$1,200**.
Savings: **\$800**.

Section 2: Quick Lesson — Simple Interest ($I = Prt$)

The Simple Interest Formula:

$$I = P \times r \times t$$

- **I** = Interest (the extra money earned or owed)
- **P** = Principal (the starting amount)
- **r** = Annual interest rate (as a decimal: 5% = 0.05)
- **t** = Time in YEARS (6 months = 0.5 years)

$$\text{Total Amount} = \text{Principal} + \text{Interest} = P + I$$

Common Mistakes:

- Forgetting to convert % to decimal (5% = 0.05, NOT 5)
- Forgetting to convert months to years (6 months = 0.5)

Worked Examples:

Example 1:

You deposit \$1,000 at 5% annual interest for 3 years. How much interest do you earn?

Solution: $I = P \times r \times t = \$1,000 \times 0.05 \times 3 = \text{\$150}$. Total = \$1,150.

Example 2:

You borrow \$5,000 at 8% for 2 years. How much interest will you owe?

Solution: $I = \$5,000 \times 0.08 \times 2 = \text{\$800}$. Total owed = \$5,800.

Example 3:

A savings account has \$2,500 at 3.5% for 4 years. Find the total amount.

Solution: $I = \$2,500 \times 0.035 \times 4 = \text{\$350}$. Total = \$2,850.

Example 4:

You invest \$10,000 at 6% for 18 months. Find the interest.

Solution: Convert: 18 months = 1.5 years. $I = \$10,000 \times 0.06 \times 1.5 = \text{\$900}$.

Example 5:

Find the interest on \$3,000 at 4% for 9 months.

Solution: Convert: 9 months = $9/12 = 0.75$ years. $I = \$3,000 \times 0.04 \times 0.75 = \text{\$90}$.

Section 3: Quick Lesson — Paycheck & Tax Math

Understanding Your Paycheck:

- **Gross Pay** = Total earnings BEFORE deductions.
- **Net Pay (Take-Home)** = What you actually receive AFTER deductions.
- **Net Pay = Gross Pay - All Deductions**

Common Deductions:

- Federal Income Tax (10%–37% depending on bracket)
- State Income Tax (varies by state)
- Social Security: 6.2% of gross pay
- Medicare: 1.45% of gross pay

Overtime: Hours over 40/week paid at 1.5× regular rate (time-and-a-half).

Worked Examples:

Example 1:

You work 40 hours at \$15/hr. What is your gross weekly pay?

Solution: Gross = $40 \times \$15 = \600 per week.

Example 2:

Gross pay is \$800/week. Federal tax is 12%, state tax is 4%. Find net pay.

Solution: Federal: \$96. State: \$32. Total deductions = \$128. Net = $\$800 - \$128 = \$672$.

Example 3:

Annual salary is \$52,000. What are monthly Social Security and Medicare deductions?

Solution: Monthly gross = \$4,333. SS: $6.2\% \times \$4,333 = \mathbf{\$268.67}$. Medicare: $1.45\% \times \$4,333 = \mathbf{\$62.83}$.

Example 4:

You work 45 hours at \$18/hr with overtime after 40 hours. Find gross pay.

Solution: Regular: $40 \times \$18 = \720 . OT rate: \$27/hr. OT pay: $5 \times \$27 = \135 .
Total = **\$855**.

Section 4: Practice Drill — 30 Real-World Money Problems

Directions: Solve each problem. Show your work. Round money to nearest cent.

Budgeting (1–8):

1. Monthly income: \$3,500. How much for Needs (50%)?
2. Monthly income: \$3,500. How much for Wants (30%)?
3. Monthly income: \$3,500. How much for Savings (20%)?
4. Annual salary: \$42,000. What is monthly take-home if total taxes are 22%?
5. You earn \$2,600/month. Rent=\$900, food=\$400, car=\$300. Are needs within 50%?
6. Monthly income: \$5,000. Using 50/30/20, how much for savings?
7. Car payment is \$350/month. Monthly income is \$2,800. What percent goes to car?
8. Your monthly budget is \$3,200. You spend \$1,500 on needs, \$1,000 on wants, \$700 on savings. What % is savings?

Simple Interest (9–18):

9. $I = Prt$. $P = \$2,000$, $r = 4\%$, $t = 5$ years. Find I .
10. $I = Prt$. $P = \$7,500$, $r = 6\%$, $t = 3$ years. Find total amount.
11. $I = Prt$. $P = \$500$, $r = 2.5\%$, $t = 2$ years. Find I .
12. You deposit \$3,000 at 4.5% for 2 years. What's the total?
13. A loan of \$12,000 at 7% for 4 years. Total interest?
14. $I = Prt$. $P = \$15,000$, $r = 5.5\%$, $t = 6$ months. Find I .
15. You save \$200/month for 12 months at 3% simple interest on the total. How much interest?
16. You invest \$5,000 at 4% for 3 years. Find total amount.
17. A car loan: \$18,000 at 5% for 5 years. Total interest?
18. You deposit \$8,000 at 3.25% for 2 years. Find the interest earned.

Paycheck & Real-World (19–30):

- 19.** Hourly rate: \$12.50. Hours: 35. Find gross pay.
- 20.** Hourly rate: \$20. Hours: 40. Find weekly gross pay.
- 21.** Gross pay: \$950/week. Federal tax: 15%. State tax: 5%. Find net pay.
- 22.** Annual salary: \$60,000. Monthly Social Security deduction (6.2%)?
- 23.** You work 50 hours at \$16/hr, overtime after 40 hours. Gross pay?
- 24.** A \$75 item has 9% sales tax. Total cost?
- 25.** Dinner bill: \$48. You tip 20%. Total paid?
- 26.** You earn \$14/hr and work 38 hours. Gross pay?
- 27.** Gross monthly: \$4,000. Total deductions = 28%. Net pay?
- 28.** You earn \$55,000/year. What's your monthly gross?
- 29.** Tip on a \$65 meal at 18%?
- 30.** Annual raise: 3% on \$45,000 salary. New salary?

Section 5: Answer Key

1. \$1,750
2. \$1,050
3. \$700
4. Monthly gross = \$3,500. After 22% tax: \$2,730
5. Needs = \$1,600. Budget = \$1,300. Over by \$300.
6. $20\% \times \$5,000 = \$1,000$
7. $350/2800 = 12.5\%$
8. $700/3200 = 21.875\%$
9. $I = \$2,000 \times 0.04 \times 5 = \400
10. $I = \$1,350$. Total = \$8,850
11. $I = \$500 \times 0.025 \times 2 = \25
12. $I = \$270$. Total = \$3,270
13. $I = \$12,000 \times 0.07 \times 4 = \$3,360$
14. $I = \$15,000 \times 0.055 \times 0.5 = \412.50
15. Total saved = \$2,400. $I = \$72$
16. $I = \$600$. Total = \$5,600
17. $I = \$18,000 \times 0.05 \times 5 = \$4,500$
18. $I = \$8,000 \times 0.0325 \times 2 = \520
19. $35 \times \$12.50 = \437.50
20. $40 \times \$20 = \800
21. Fed: \$142.50, State: \$47.50. Net = \$760
22. Monthly = \$5,000. SS = \$310
23. Reg: $40 \times \$16 = \640 . OT: $10 \times \$24 = \240 . Gross = \$880
24. $\$75 \times 1.09 = \81.75
25. $\$48 \times 1.20 = \57.60

26. $38 \times \$14 = \532

27. $\$4,000 \times 0.72 = \$2,880$

28. $\$55,000 \div 12 = \$4,583.33$

29. $18\% \times \$65 = \11.70

30. $3\% \times \$45,000 = \$1,350$. New = \$46,350

